# Session 10
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Module 2, Week 4

# The Pandas DataFrame Essentials

Importing external datasets; Pandas; The Pandas DataFrame; Importing data; Initial data inspection; Selection

# What and why NumPy

- Before training ML models, data usually needs:
  - *Storage, Preprocessing and Feature Extraction*
- Pandas helps perform all these tasks efficiently on the entire data.

*CSV File → Pandas DataFrame → Data Cleaning → ML Model*

DataFrames help:

- Store structured data
- Analyze datasets
- Filter and sort records
- Prepare data for ML models

Used in:

- Data Science
- Machine Learning
- Research
- Business Analytics

## Importing Pandas library in Python and a Dataframe

*import pandas as pd*

*print("Pandas imported successfully!")*

- A DataFrame in Pandas is a table-like data structure with rows and columns like excel and csv.
  - Keys become column names
  - Lists become column values

```
data = {
"Name": ["Aman", "Riya", "Kunal"],
"Age": [20, 21, 19],
"Marks": [85, 90, 78]
}
```

*#Creating a dataframe*

```
df = pd.DataFrame(data)
print(df)
```

```
   Name  Age  Marks
0 Aman   20   85
1 Riya   21   90
2 Kunal 19    78
```

## NumPy and/Vs Pandas

| Library | | Purpose |
|---|---|---|
| NumPy | - | Numerical arrays and computations |
| Pandas | - | Tabular data analysis |

Pandas internally uses NumPy arrays for fast computation.

$Pip install numpy

$pip install pandas

# Pandas: Accessing a Dataframe

```
data = {
"Name": ["Aman", "Riya", "Kunal"],
"Age": [20, 21, 19],
"Marks": [85, 90, 78]
}
```

```
df = pd.DataFrame(data)
print(df)
```

| | Name | Age | Marks |
|---|---|---|---|
| 0 | Aman | 20 | 85 |
| 1 | Riya | 21 | 90 |
| 2 | Kunal | 19 | 78 |

**Accessing Columns**

print(df[["Name", "Marks"]])

| | Name | Marks |
|---|---|---|
| 0 | Aman | 85 |
| 1 | Riya | 90 |
| 2 | Kunal | 78 |

**Accessing rows**

df.loc[0]

Print row at index 0

Df.loc[0:2]

Print rows from 0 to 2

# Pandas: Objects to get the shape of a dataframe

Shape of a dataframe: shape

print(df.shape) -> (3,3)

print(df.columns) -> Index(['Name', 'Age', 'Marks'], dtype='object')

df.columns.tolist()  -> ['Name', 'Age', 'Marks']

# Read and write CSV functions

- Create the CSV file named as students.csv

```
df.to_csv("students.csv", index=False)

#index=False prevents Pandas from
saving row numbers.
```

- Read the CSV

```
new_df = pd.read_csv("students.csv")

print(new_df)

     Name  Age Marks
0    Aman  20  85
1    Riya  21  90
2    Kunal 19  78
```

# Inspection of the data in dataframe

Get n number of rows from head of the table: *df.head(n)*  # default is first 5 rows

Get n number of rows from tail of the table: *df.tail(n)* # default is last 5 rows

Get complete info (number of columns, rows, instances, data type in each row, etc) of the table: *df.info()*

*#The object datatype in Pandas generally represents textual or mixed-type data stored as Python objects.*

Get statistics related to numerical columns in the dataframe: *df.describe()*

*Count of values, Mean, Standard deviation, Minimum value, Quartiles (25%, 50%, 75%), Maximum value*

# Data Analysis using Filtering

- Filtering: Selecting only the rows that satisfy certain conditions.

Select students with marks greater than 80

```
filtered_df = df[df["Marks"] > 80]

print(filtered_df) ->

   Name  Age Marks

0  Aman  20  85
1  Riya  21  90
```

# Data Analysis for Missing Values

```
import pandas as pd

import numpy as np

data = {

    "Name": ["Aman", "Riya", None, "Neha"],

    "Marks": [85, np.nan, 76, 90],

    "City": ["Delhi", "Mumbai", None, "Pune"]

}

df = pd.DataFrame(data)
```

```
print(df)

Name Marks City
0 Aman 85.0 Delhi
1 Riya NaN Mumbai
2 None 76.0 None
3 Neha 90.0 Pune
```

```
print(df.isnull())

  Name Marks City
0 False False False
1 False True False
2 True False True
3 False False False
```

```
print(df.isnull().sum())
Name 1
Marks 1
City 1
-----------------------------
print(df.isnull().sum().sum())  -> 3
```

# Thank you!